Multi-Temporal Resolution-Guided 3-D Imaging for Array-Based Single-Photon LiDAR

Kai Qiao, Yu Chang, Zefang Xu, Fei Yin, Chang Su, Liyu Liu, Tianye Liu, Chunliang Liu, Jinshou Tian, and Xing Wang

***Abstract*—Using a 1550 nm array-based single-photon LiDAR system, we demonstrated depth profiling of both static and dynamic targets up to a distance of 10 kilometers. The system comprises a 1550 nm pulsed laser source, a bistatic optical transceiver system, and a 64 × 64 InGaAs/InP Single Photon Avalanche Diode (SPAD) array camera, with an angular resolution of 20 μrad. By employing a recovery optimization algorithm guided by multi-scale time resolution, we utilized unsupervised learning methods to achieve three-dimensional (3-D) image segmentation. Subsequently, we accomplished pixel-level algorithm matching, facilitating efficient long-range 3-D imaging reconstruction with significantly reduced binary frame data. Notably, after offline processing of real point cloud data collected by our system, we obtained depth images of various targets within a 4 to 10 km range. Furthermore, we successfully captured dynamic 3-D video of targets at a frame rate exceeding 50 fps. The video was reconstructed using offline processing with an average of fewer than 2 photons returned per pixel. These depth results highlight the potential of the proposed system and reconstructed method for depth profiling, feature extraction, and target recognition of distant static and dynamic targets.**

***Index Terms*—Single-photon detector, three-dimensional video, LiDAR.**

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I. Introduction

SINGLE-PHOTON light detection and ranging (LiDAR), which provides high-resolution 3-D images, has been successfully validated in challenging scenarios such as long-range depth imaging [1], non-line-of-sight imaging [2], and global terrain imaging [3], owing to its unique single-photon sensitivity and excellent high temporal resolution [4].

The use of a 1550-nm wavelength in single-photon LiDAR systems reduced solar background interference, decreased atmospheric losses, and improved eye safety thresholds, enabling the use of higher optical power levels [5], [6], [7]. The remarkable sensitivity and compactness of the InGaAs/InP SPAD detector [8] facilitate weight reduction and power efficiency in single-photon LiDAR systems, making them exceptionally suitable for integration into a wide range of platforms, such as aerial surveillance [9] and satellite-based mapping [10].

Researchers have conducted studies on LiDAR systems and 3-D image reconstruction algorithms based on single-element InGaAs/InP SPAD detector, confirming its advantages for long-range 3-D imaging [11]. Each pixel requires acquisition times ranging from milliseconds to hundreds of milliseconds; as a result, this approach significantly limits the scanning frame rate and echo density. The use of SPAD detector array [12] has the potential to reduce the acquisition time for complete images and simplify the optical configuration, thereby enhancing overall efficiency, which has demonstrated significant application potential in various fields such as security defense, autonomous driving [13], planetary remote sensing [14] and marine engineering [15], [16], [17], leading to a growing interest among researchers.

In recent research, Tachella et al. [18] demonstrated a video rate of 50 fps with a processing time of 20 ms when using offline data from InGaAs/InP SPAD detector array camera, which is the first example of real-time processing for multi-surface single-photon data. Clearly, to achieve video of 3-D imaging, it is necessary to combine a SPAD detector array with fast image processing algorithms. Besides improving algorithm processing speed, utilizing as few binary frame data as possible through

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effective 3-D reconstruction algorithms is another method to increase imaging speed, which has become one of the hotspots in the research of single-photon LiDAR. Jiang et al. [19] developed a single-photon LiDAR based on a 64 × 64 InGaAs/InP array. They also carried out the corresponding optical design and optimized the reconstruction algorithms. In the experiment, they collected scene data with a total acquisition time of 2 s, during which 50 k binary frames were captured. Then, they combined 1250 binary frames (corresponding to a 50 ms integration time) in sequence for depth map reconstruction. After reconstruction using the optimized algorithm, a video was achieved, showing the wind turbine rotation at 20 frames per second (fps) from a distance of 10 km. However, the application of array based single-photon LiDAR in achieving long-distance, high-speed 3-D imaging still faces numerous challenges. Long-distance detection and the low signal-to-noise ratio (SNR) resulting from the use of effective binary frames, as well as the broadening of returned light in rough terrain or dynamic applications, and the additional noise introduced by optical crosstalk between pixels, which can cause echo photons to be submerged in the noise, all contribute to the loss of target depth information.

In this paper, we present hardware and offline algorithm techniques specifically designed for remote scenes, which are employed to tackle the intricate challenges associated with long-distance area-array active imaging. In terms of hardware, we constructed a visible-near-infrared co-optical axis system that employs a large optical aperture and narrowband filters. These are designed to enhance photon acceptance efficiency and reduce background noise. In the domain of algorithms, we have developed an efficient approach for long-range 3-D imaging that harnesses multi-scale time resolution. This method constructs correlation coefficients between depth maps of varying temporal resolutions, which are directly related to the target's reflectivity and distance, by utilizing unsupervised learning methods to achieve precise pixel-level segmentation. To reconstruct distances for individual pixels, we employ adaptable matching techniques [20], [21]. These improvements can achieve 3-D imaging of complex scenes at distances of 4-10 km in real outdoor conditions, with an angular resolution of 20 μrad and fewer than 2 photons per pixel (ppp). In addition, we have achieved offline processing for 3-D imaging of moving targets at a distance of 10 kilometers at 50 fps, which is of significant importance for the development of a truly real-time 3-D imaging single-photon LiDAR system.

II. Experimental Setup and Reconstruction Algorithm

The single-photon LiDAR system outlined in this paper primarily comprises an optical receiving system (ORS), a SPAD array camera, a laser, and a CCD camera, as depicted in Fig. 1(a). In our system, we have integrated an InGaAs/InP 64 × 64 SPAD detector array camera with an effective photosensitive area of 3.2 mm × 3.2 mm. This camera can simultaneously receive and process echo photons from targets within a specific area, reducing data acquisition time by thousands of times and effectively increasing the frame rate of 3-D imaging compared to a single-element SPAD detector operating in scanning mode.

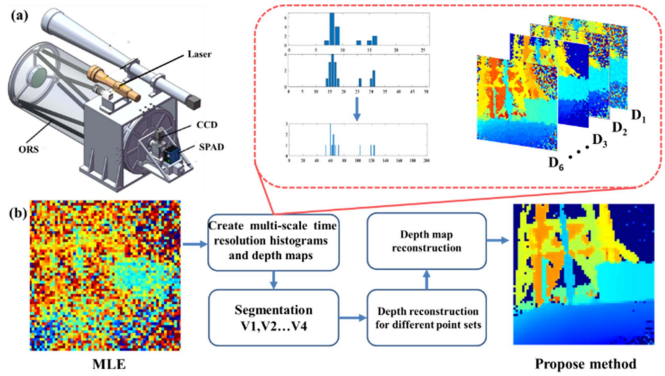


Fig. 1. (a) Overall design of the single-photon LiDAR system. (b) Workflow diagram showing the process of: generating multi-scale time resolution histograms and depth maps, the correlation between distance images with varying temporal resolutions to achieve pixel segmentation, obtaining reconstructed depth through the application of diverse methods, completing the depth map reconstruction.

TABLE I
Summary of the System Parameters

Parameter	Value
Wavelength	1550 nm
Maximum average power	1.25 W
Pulse width	1ns
Divergence angle	1.5 mrad
Pulse repetition frequency	25 kHz
Photon detection efficiency	20%
Detector array	64 × 64
Time bin	1 ns
Focal length f	3000 mm
Optical aperture l	300 mm
Narrow band filter width	4nm
Field of view	1.2 mrad

The laser source and the receiving optical system adopt a bistatic optical configuration, which effectively suppresses backscattered noise from local optical components and the near-field atmosphere.

The ORS utilizes a designed Cassegrain optical telescope with a 300-mm effective optical aperture, a 3000-mm focal length, and a total effective field of view (total FOV) of 1.2 mrad. A bandpass filter centered at 1550.0 nm, with a full width at half maximum of 4.0 nm, is adopted. This filtering design provides adequate suppression of background noise in the atmosphere, enabling the system to operate during certain periods of the day, even in daylight. An all-fiber pulsed erbium-doped fiber laser is selected as the system's light source, with a working wavelength of 1550 nm (pulse width of 1 ns, repetition frequency of 25 kHz), and a maximum average laser power of up to 1.25 W. The laser light is transmitted through the fiber to a collimator and then illuminates the scene with a divergence angle of 1.5 mrad, slightly larger than the system's total FOV.

The main parameters of the proposed system are given in Table I.

Recently, a global gating approach and an inverse 3-D deconvolution process based on the modified SPIRALTAP solver have been proposed to solve the multiple-returns problem in a long-range situation, which is caused by the divergence of the laser beam and the receiver's large FOV [19], [22]. To implement photon-sensitive detection with an extremely low signal-to-background ratio (SBR), a single-photon imaging mechanism utilizing a multi-scale histogram is proposed [21]. This mechanism utilizes the central pixel and its surrounding eight pixels to form a pixel block. Due to the spatial correlation among these pixel blocks, echo photons can be clustered into time bins within the macro-histogram of the pixel blocks, thereby separating them from noise. However, obtaining a macro-pixel histogram by leveraging inter-pixel correlation cannot suppress cross-talk signals between pixels. D Shinet al. formulated a convex optimization problem by employing discretization and relaxation techniques. Subsequently, they utilized an enhanced iterative shrinkage-thresholding algorithm to swiftly compute the optimal solution for multiple depth levels, thereby recovering the missing information [23]. These methods apply image-level processing or global noise reduction techniques to pixels, therefore, effectively separating the signal from noise remains challenging for local pixel points, leading to the loss of local image information and a deterioration in image resolution.

To significantly increase the efficiency of 3-D scene reconstruction for array-based single-photon LiDAR by utilizing fewer binary frames for real-world long-range imaging is our primary objective in this work. In the context of data processing, to mitigate interference among pixels and information loss in local regions within the area array device, we propose an efficient pixel-level 3-D reconstruction method guided by multi-scale time resolution. This method enables the acquisition of depth maps using sparse photons in complex scenarios.

A. Observation Model

In photon-starved regimes, the observed laser counting rate of the system falls well below the theoretical prediction based on the laser repetition rate. Consequently, the photon counting statistics in follow-up experiments adhere to a Poisson distribution. We can formulate the count of photons recorded by the SPAD at the timestamp index n and the pixel position (j,k) in the array as

$$N(j,k)[n]=\int_{(n)\Delta t}^{(n+1)\Delta t} R\,(j,k)\,s\left(t-\frac{2D(j,k)}{c}\right)dt+N_{noise}. \tag{1}$$

Here, the notation $N(j,k)[n]$ represents the quantity of echo photons captured during the time-stamp interval n. The parameter Δt corresponds to the duration of each time bin within the Time to Digital Converter. $R(j,k)$ signifies the reflectivity characteristics of the scene under observation, while $D(j,k)$ indicates the spatial separation between the target object and the system. The N_{noise} represents the background noise. The s denotes the waveform of a single pulse. In accordance with the principles of LiDAR, both a lower target reflectivity and a greater distance will result in a decrease in the number of echo photons.

Given that each detected photon operates independently of the others, the photon count distribution across N illumination intervals adheres to a Poisson distribution. Consequently, the temporal histogram representing photon arrivals can be expressed through the following formulation:

$$H(j,k) \sim Poisson\{n(N(j,k)[n]+N_d)\}. \tag{2}$$

where $H(j,k)$ denotes the histogram that records the photon counts corresponding to the imaging pixel located at coordinates (j, k). Meanwhile, N_d signifies the rate at which dark counts are registered.

In rough terrain or dynamic application scenarios, the echo optical pulse may undergo broadening. Under such circumstances, the echo photons may be discretely distributed across multiple time bins in a histogram, causing them to be submerged in the noise [24]. At this point, it is necessary to degrade the time resolution of the histogram to cluster the echo photons into a single time bin. Unfortunately, this degradation in time resolution may not only lead to a decrease in depth measurement accuracy but also result in a corresponding increase in the average number of noise photons.

B. Depth Reconstruction

The reconstruction, illustrated in Fig. 1(b), contains three steps:

i. Multi-scale time resolution histograms for each pixel are generated, with temporal resolutions ranging from 1 ns to 32 ns. In this part, A basic histogram is generated by conducting pixel-by-pixel statistical analysis (with atemporal resolution of 1 ns). Subsequently, adjacent bins are merged in groups of 2, 4, 8, 16, and 32, respectively, to produce histograms with time resolutions of 2 ns, 4 ns, 8 ns, 16 ns, and 32 ns. These histograms are produced to cluster the echo photons into a time bin and separate them from the noise.

The histogram with various time resolutions undergo a discrete pixel-level cross-correlation operation, (IRF∗H)[n], between the recorded histogram H[n] (with N time bins) and the instrument response function (IRF), as described below

$$(IRF*H)[m]=\sum_{m=0}^{m=M} IRF[n]H[n+m]. \tag{3}$$

$$d=\frac{rag_{m\in m_r}^{\max}(IRF*H)[m]}{2}\times c. \tag{4}$$

where m represents the displacement magnitude. The target is situated precisely at the histogram's center, and cross-correlation computations are performed by considering displacement values that span the interval $m_r = [0 + M/2, N - M/2]$. This approach is intended to minimize computational errors that arise when the peak of the probability density function is located at or near the edges of the histogram H[t]. The depth d (measured in the number of time bins), which corresponds to each pixel's relative distance from the target, is given as the highest cross-correlation value. Depth maps D_1, D_2, …, D_6 with temporal resolutions ranging from 1 ns to 32 ns are obtained. These depth images were

estimated from the various temporal histograms of the SPAD camera using (3) and (4).

ii. Subsequently, the correlation coefficient is calculated between the depth values in depth map D_6 and the depth maps with other time resolutions for the same pixels, yielding a correlation coefficient map C. The correlation coefficient map C can be computed as

$$C(j,k) = \frac{1}{\frac{1}{5}\sum_{i=1}^{5} |D_6(j,k) - D_i(j,k)|}. \tag{5}$$

In real - world 3-D imaging, the reflectivity and distance of targets vary significantly across different regions in the system's FOV. Even though uniform laser illumination is used to cover the imaging area, according to the single - photon lidar equation, there are substantial differences in the number of echo photons corresponding to each pixel in the array. Based on the correlation coefficient map, all pixels can be classified into the following categories:

a) The first category encompasses pixels that have acquired valid distance values from depth images with varying temporal resolutions and demonstrate high correlation coefficients, indicating a high SBR for the corresponding pixels.
b) The second category comprises pixels whose depth values fluctuate randomly across different temporal resolutions, corresponding to low correlation coefficients. These pixels are regarded as background pixels, representing regions in the image where no echo photon is present.
c) The third category includes pixels that exhibit a low SBR, resulting in correlation coefficients that are distributed over a wide range across different temporal resolutions. During the depth reconstruction phase, it is challenging to obtain distance information for all pixels using global or unified algorithms, leading to missing or erroneous information for some pixels. In subsequent clustering processing, reasonable thresholds will be designed to further categorize the data into two distinct point sets.

In the next step, the fuzzy c-means clustering (FCM) segmentation algorithm, constrained by the intensity image, is applied to segment the correlation coefficient map C. FCM stands out as a highly prevalent fuzzy clustering algorithm, renowned for its remarkable performance in both the realms of data processing and image segmentation tasks. It integrates the theory of fuzzy sets into the clustering process, enabling each data sample to possess a varying degree of membership across multiple clusters. This approach ensures that the clustering outcomes more closely mirror the complexities and nuances found in real-world situations [25], [26], [27].

Reshape the correlation coefficient map C into dataset as $X = [X_1, X_2, \ldots X_{M\cdot N}]$, the fuzzy C-means method aims to partition the dataset into 'c' distinct clusters, with the primary goal of minimizing the associated loss function. The formulation of the loss function for fuzzy C-means, along with its corresponding constraints, is presented as follows:

$$\min \sum_{i=1}^{M\cdot N}\sum_{j=1}^{c} \| X_i - v_j \|_2^2 \, u_{ij}^m,$$
$$\text{s.t.} U \geq 0, \sum_{j=1}^{c} u_{ij} = 1. \tag{6}$$

$V = \{v_1, \cdots, v_k, \cdots, v_c\}$ in (6) represent c cluster centers v_k. c is the number of clusters. The $\|X_i - v_j\|_2^2$ represents the squared Euclidean distance between the *i*-th data point and the *j*-th cluster center. Meanwhile, u_{ij} signifies the degree of association or membership of the *i*-th sample with respect to the *j*-th cluster center. Additionally, there is a constraint ensuring that the cumulative membership of each sample across all clusters sums up to unity.

By applying the Lagrange multiplier technique to address (6) while adhering to the given constraint, we derive the subsequent updating formulas for the matrices U and V:

$$u_{ij} = \frac{1}{\sum_f^c \left(\frac{\| x_i - v_j \|_2^2}{\| x_i - v_f \|_2^2} \right)^{\frac{1}{m-1}}}, 1 \leq i \leq M\cdot N, 1 \leq f \leq c,$$
$$v_j = \frac{\sum_{i=1}^{M\cdot N} x_i u_{ij}^m}{\sum_{i=1}^{M\cdot N} u_{ij}^m}, 1 \leq j \leq c. \tag{7}$$

Reshape the photon count map into a dataset denoted as $G = [G_1, G_2, \ldots, G_{M\cdot N}]$. The $f(G_i)$ represents the normalized photon count of the *i*-th pixel, with $f(G_i) = G_i / \max(G)$, where max(G) is the maximum value in the photon count image. The fuzzy membership U is updated using (8), and the new fuzzy memberships u''_{ij} and cluster centers v'_j are obtained as follows

$$u'_{ij} = \frac{a \cdot u_{ij} + b \cdot f(G_i)}{a + b}, 1 \leq i \leq M\cdot N, 1 \leq j \leq c,$$
$$u''_{ij} = \frac{u'_{ij}}{\sum_c u'_{ij}},$$
$$v'_j = \frac{\sum_{i=1}^{M\cdot N} x_i u''^{\,m}_{ij}}{\sum_{i=1}^{M\cdot N} u''^{\,m}_{ij}}, 1 \leq j \leq c. \tag{8}$$

when $\max |u''_{ij}(t+1) - u''_{ij}(t)| < \varepsilon$, the program will stop iterating, where t represents the total count of iterations completed, and ε denotes the predefined stopping threshold, which is set within the interval [0,1].

iii. In the last step, different methods are employed to calculate the depth information corresponding to different pixels. For the first point set, the peak-finding method is utilized for depth reconstruction. A constant value is assigned to the second point set because it has not received any echo signals. For the third point set, guided by the distance information obtained from pixels with lower temporal resolutions, the cross-correlation algorithm is utilized within a defined distance range [D_6 - τ/2, D_6 + τ/2] for distance reconstruction. Here τ represents the gating width. For the fourth set, we introduce Total Variation (TV) and the sparseness regularization terms of Discrete Cosine

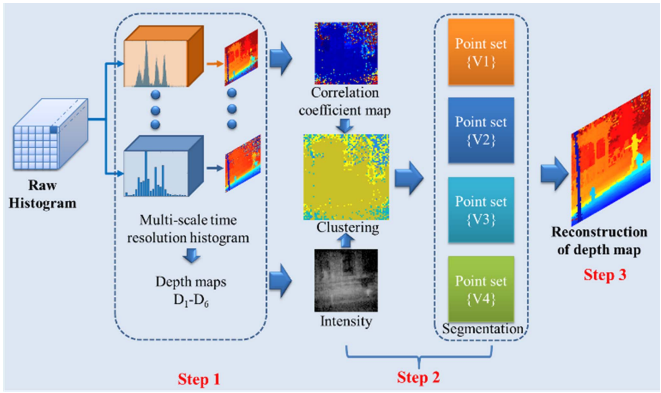


Fig. 2. Schematic of 3-D imaging algorithm for single-photon LiDAR based on multi-scale temporal resolution guidance.

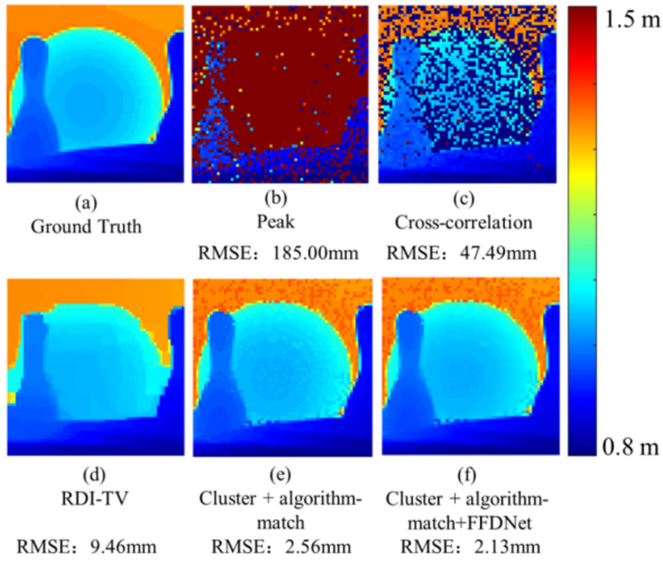


Fig. 3. Deep reconstruction effect. (a) The true depth map of the object. (b) The pointwise peak-finding depth estimation. (c) The pointwise cross-correlation depth estimation. (d) The RDI-TV depth estimation. (e) Reconstruction results obtained by using cluster-based segmentation and the dynamic algorithm matching. (f) The reconstructed result obtained by applying FFDNet denoising to (e).

Transform (DCT) coefficients, and employ the Alternating Direction Method of Multipliers (ADMM) to solve the convex optimization problem [11]. This approach is highly suitable for depth profile restoration from sparse single-photon data, facilitating the recovery of corrupted depth and intensity images as well as the reconstruction of missing pixels. It has been applied in both underwater imaging and free-space scenarios.

Finally, the depth values obtained for pixels from different point sets are fused based on their corresponding pixel positions to generate the final depth map D. The obtained depth map is subjected to smoothing using the neural network-based FFDNet algorithm [28], resulting in a clearer and more accurate image.

The principle of efficient long-range 3-D imaging reconstruction for single-photon guided by multi-scale temporal resolution depth maps is depicted in Fig. 2.

III. Experimental Results and Discussion

A. Simulation Experimental Results

To evaluate the performance of our proposed depth map reconstruction method, we conducted two types of simulation experiments under low SBR conditions. In the first simulation experiment, an ablation study was performed on the proposed algorithm to verify that, after incorporating cluster-based segmentation, dynamic algorithm matching, and denoising, the reconstructed images exhibit superior distance accuracy compared to those obtained using any single algorithm among the peak-finding method, cross-correlation method, and RDI-TV. The second simulation conducted a comparative analysis of the depth image reconstruction performance among maximum likelihood estimation (MLE), the unmixing algorithm proposed by Rapp [29], a long-range-tailored computational algorithm developed by Li [23], and the algorithm presented in this work, under varying PPP and SBR conditions. The objective was to verify that our method demonstrates superior performance under low SBR conditions.

In order to meet the low SBR requirement, the number of laser pulse illuminations is set to 100. The average photon detection rate was one detection in approximately 20% of the illumination periods. The effective detection time for each pixel is T = 2000 ns. The external detection condition parameters are set as follows: the PPP values are 1.0 and 2.0, and the SBR value is 0.2.

The reconstruction quality of the depth estimation image is evaluated using the root mean square error (RMSE) between the true value of the depth map (generated by simulation) and the reconstruction result. The RMSE can be formulated as

$$\mathrm{RMSE}\left(\hat{Z}, Z\right) = \sqrt{\frac{1}{m \times n} \sum_{x=1}^{m} \sum_{y=1}^{n} \left(Z_{x,y} - \hat{Z}_{x,y}\right)^2}. \tag{9}$$

Upon comparing Fig. 3(b) and (c) with Fig. 3(a), it can be observed that both the point-by-point peak-finding method and the cross-correlation method perform poorly and are sensitive to low PPP levels and the presence of background noise. However, the latter can better preserve details such as the contours of the measured object. Although RD-TVI demonstrates the ability to remove most of the noise covering the entire image and yields better RMSE, it is evident that the resulting depth maps are blurred as shown in Fig. 3(d). However, a comparison among Fig. 3(e), (f), and (a) reveals that by introducing cluster-based segmentation, the dynamic algorithm matching can not only suppress noise but also clearly discern the contours of the target, thereby obtaining more accurate depth information. Furthermore, employing the FFDNet denoising module removes most of the noise outside the background region and enables the finally reconstructed image to restore the true depth of the measured object to the greatest possible extent. This conclusion can also be validated by the values of the RMSE.

Fig. 4 presents the exemplary simulated results for two scenarios under PPP conditions of 1.0 and 2.0, respectively, with an SBR value of 0.2. The original depth map is provided as a reference to illustrate the outcomes processed by different methods

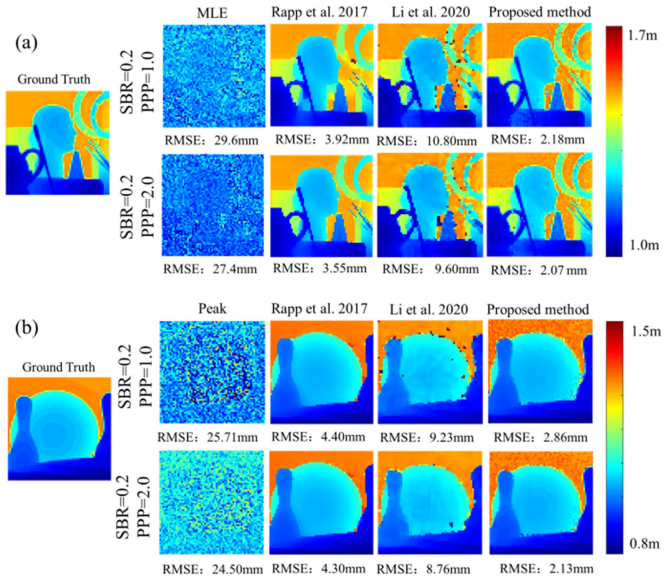


Fig. 4. Simulated processing results for Art and Bowling scenes at 1.0 and 2.0 PPP, and SBR fixed to 0.2.

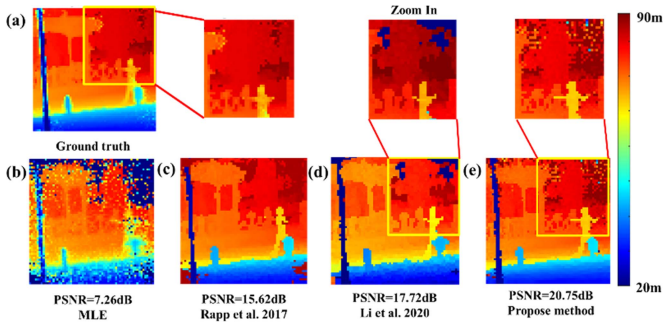


Fig. 5. Depth map of a person in a complex background, obtained from experimental measurements at a distance of 4.0 km. (a) The reconstructed depth results with a total data acquisition time of 0.8s. (b)–(e) Reconstruction results obtained by using the pixel wise MLE, unmixing algorithm, long-range-tailored computational algorithm by Li, and the proposed algorithm, respectively. The single-photon Lidar recorded an average PPP of 6.60, and the SBR was 0.26. Note that the enlarged images showcase depth maps of local details and distant targets.

from noisy, low-signal-photon data. We compared the results of our proposed method with those from the MLE approach, as well as the methods developed by Rapp et al. and Li et al. In the case of depth estimation, the MLE method fails because noise detection tends to pull the estimated depth values toward the mean depth of the scene. As shown in Fig. 4, although our method does not demonstrate superior denoising performance, it is capable of identifying the contours of the target and obtaining more accurate depth information. As depicted in Fig. 4(a), our method shows marked performance improvements in terms of RMSE compared to the methods put forward by Rapp and Li, with RMSE reductions amounting to 41.6% and 78.1%, respectively.

B. Real-World Experimental Results

In this paper, different targets at varying distances were carefully selected, taking into comprehensive consideration their spatial layout, reflectivity, and structural complexity, to fully exhibit 3-D imaging capabilities of our system and algorithm. Specifically, we chose a pedestrian and a pylon situated at 4 km and 10 km respectively, as well as a wind turbine located 10 km away, as our observation subjects. During the experimentation process, the system's FOV was stably maintained at 1.2 mrad and the angular resolution was kept constant at 20 μrad, unaffected by the target's distance. For instance, this angular resolution corresponds to the ability to discern a target approximately 20 cm in size at a distance of 10 km. In the ensuing sections, we will present image examples captured of individuals within complex structures, a distributed target, and a dynamic imaging at 50 fps for rotating wind turbine. The single-photon LiDAR system proposed in this paper is utilized to perform 3-D imaging with a total acquisition time of 5 seconds. The reconstructed depth map, obtained using the MLE method, serves as the ground truth in all experiments.

1) Person in a Complex Background At 4 km: A house located 4 km away was selected as the imaging area, and a person was added to the scene as the imaging target. The proposed LiDAR system was used to detect this area, outputting a 64×64 pixels image with a spatial resolution of approximately 20 cm and a total acquisition time of 0.02 s (500 binary frames for the proposed system). The averaged PPP was $\sim$ 6.60, were calculated based on all the pixels in the entire scene. The quantitative performances are given with peak-signal-to-noise-ratio (PSNR) while the ground truth is chosen as the data with long data acquisition time shown in Fig. 5(a).

The plots in Fig. 5(b)–(e) show the depth maps obtained by using various reconstructed algorithms [13], [14], including MLE, unmixing algorithm by Rapp, a long-range-tailored computational algorithm by Li, and the method proposed herein. The proposed method recovers the fine details of scenes, enabling the precise identification of the house's outline as well as the size and distance information of windows and fences. However, the MLE method, when used for data reconstruction, clearly suffer from a low number of photons and noise, resulting in significant noise in the depth maps and the loss of distance information in the scene's edge regions. The unmixing algorithm and long-range-tailored computational algorithm can effectively restore the distance information of the scene, but it leads to the loss of some information; for instance, in the depth maps, the limbs and torso of a person are not clearly visible, and the spatial information of the x-shaped railing in the background has been lost. Due to the scarcity of returning photons, the depth information of targets in the far part of the scene is lost. However, by utilizing the algorithm we proposed, the distance information of targets such as distant trees is still well reconstructed, as depicted in Fig. 5(e). The PSNR of the proposed algorithm is 10.59 dB better than that of the MLE method, 5.13 dB better than that of the unmixing algorithm, and 3.03 dB better than of the long-range-tailored computational algorithm.

2) Distributed Target (a Pylon): The visible-band photograph of a pylon at a distance of 10 km, captured using a CCD camera, is depicted in Fig. 6(a). A 64×64 pixels image of the

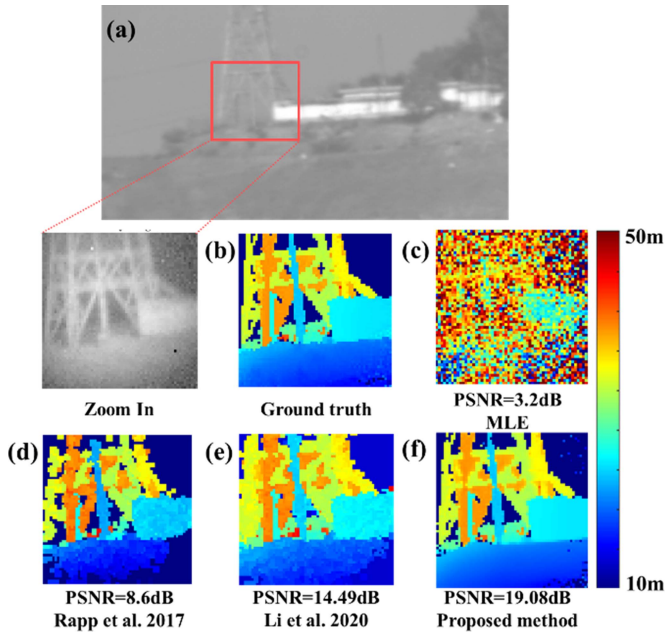


Fig. 6. Depth map of a distributed target, obtained from experimental measurements at a distance of 10.0 km. (a) Visible-band image of the target taken by a standard CCD camera mounted on the imaging system. (b) The reconstructed depth result, with a total data acquisition time of 5.0s, was chosen as the ground truth. (c)–(f) Reconstruction results obtained by using the MLE method, the unmixing algorithm, long-range-tailored computational algorithm, and the proposed algorithm, respectively. The single-photon Lidar recorded an average PPP of 2.0, and the SBR was 0.28.

pylon was obtained with a spatial resolution of 20 cm. Fig. 6(b) displays the depth map of the pylon, representing data collected over a longer acquisition time.

The plots in Fig. 6(c) to (f) display the reconstructed depth achieved through the use of various imaging algorithms. The data was collected over an acquisition time of 0.02 seconds, with an PPP of approximately 2.0 and a SBR of 0.28, which is defined as the ratio between the number of photons reflected from the target and the total number of counts from the background noise. Our proposed method enables us to clearly distinguish the intricate features of the structure, and it also captures depth information in the edge regions of the scene where there are fewer echo photons. Nonetheless, a significant number of data points attributed to noise are also evident. The MLE and unmixing algorithms, however, fail in this regard. The long-range-tailored computational algorithm can effectively restore the three-dimensional information of the target's partial region, revealing the overall contour of the wires and distance information for some areas, while noise in the data is well suppressed. However, due to the reduced image resolution, detailed information within the target is not displayed. Simultaneously, regions with fewer echo photons at the scene edges fail to reconstruct valid three-dimensional information. The PSNR of the proposed algorithm is 15.88 dB better than that of the MLE method, 10.48 dB better than that of the unmixing algorithm, and 4.59 dB better than of the long-range-tailored computational algorithm.

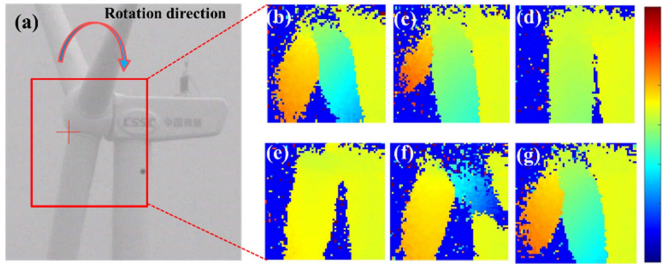


Fig. 7. Depth map of a dynamic target, obtained from experimental measurements at a distance of 10.0 km. (a) Visible-band image of the target taken by a standard CCD camera mounted on the imaging system. (b)-(g) Depth maps of various frames. Each image is captured with 20 ms data acquisition time while the time interval between the six frames is 100 ms.

3) Dynamic Target (a Rotating Wind Turbine): Benefiting from the use of the SPAD array detector and efficient depth reconstruction algorithms, our LiDAR system can also image dynamic targets at long distances within short acquisition times. In this study, our LiDAR system was able to cover an imaging area of 12 m $\times$ 12 m with a 64 $\times$ 64 array detectors, achieving a spatial resolution of approximately 20 cm. Fig. 7(a) shows a visible-band photograph of a pylon at a distance of 10 km, taken with a camera lens with a focal length of 3000 mm. The scene data was collected by our LiDAR system with a total acquisition time of 0.8 seconds (20,000 frames). During the depth reconstruction phase utilizing offline data, we combined 500 binary frames (over 20 ms) to reconstruct the depth map, enabling us to obtain continuous depth images of the rotating wind turbine. The PPP is 1.7, and the SBR is 0.4. The employment of the proposed algorithm results in the reconstruction of a video showing the rotation of the wind turbine at up to 50 fps.

As depicted in Fig. 7(b) to (g), six frames were extracted from the video at intervals of 6 frames. Despite capturing only a portion of the wind turbine area due to the limited FOV of the system, the results still clearly distinguish between the structures of the wind turbine, such as the tower and the blades. Furthermore, the periodic rotation of the blades can be observed from multiple frames of depth maps. It is evident that there is an angle between the plane of the blades and the direction of laser incidence, and that the blades of the wind turbine are rotating in a clockwise direction.

IV. Conclusion

In summary, we have demonstrated the potential of a 1550-nm array based single-photon LiDAR for achieving 3-D imaging with fewer binary frames over a range of 10 km. By utilizing a 64 $\times$ 64 SPAD detector array and incorporating an optimized reconstruction method, our system has achieved a hundredfold increase in data acquisition efficiency compared to single-element single-photon LiDAR. Based on the 3-D point cloud data of various targets acquired from a range of 4 km to 10 km through the actual system, we have obtained depth maps with acquisition times less than 20 ms via offline processing algorithms. Moreover, the employment of the proposed method results in the reconstruction of a video showing the rotation

of the wind turbine at up to 50 fps over 10 km. The proposed high - efficiency 3-D data reconstruction method has been used to generate high - quality depth maps with less binary data, and the reconstructed results have exhibited certain advantages in target recognition and confirmation, thereby promoting the development of long-range array based single-photon LiDAR imaging.

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